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Securing the Digital Mine with a Metaheuristic-Optimized Deep Learning Adaptive System

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Introduction

• Mining 4.0 & Smart Subsoil: African and Russian mineral extraction complexes are rapidly integrating IoT sensors, SCADA telemetry, and automated milling fleets to maximize ore recovery.

• The Air-Gap Myth: Connecting OT networks to enterprise analytics exposes unauthenticated Modbus RTU/TCP and DNP3 industrial protocols to hostile zero-day cyber intrusions.

• High Downtime Costs: Unplanned mining downtime costs between USD $50,000 and $500,000 per hour; cyber-physical disruption of ventilation or dewatering directly endangers human life.

• Failure of IT-Centric IDS: Enterprise security tools analyze 41+ features taking 150+ ms, directly violating the 20 to 50 millisecond control loop deadlines of industrial PLCs.

Method

Our approach is built around four interconnected modules designed to operate in real-time on edge gateways:

• **Data Ingestion & Extraction:** Promiscuously captures raw network frames from SCADA interfaces at line speed using @mhiskall282/unesco-mine-sec-cli, parsing IP/TCP/Modbus packet headers without dropping packets.

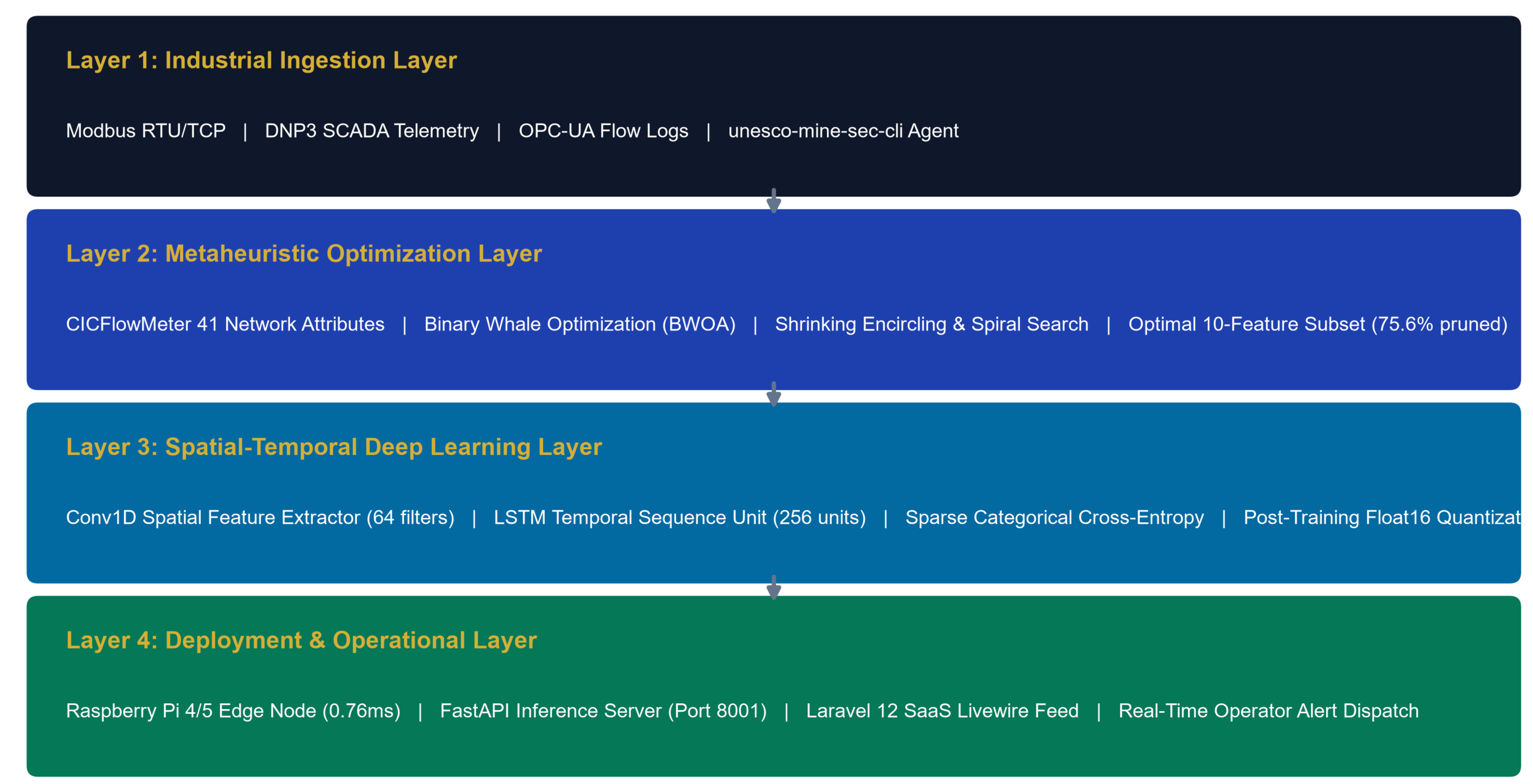
• **BWOA Feature Pruning:** Applies Binary Whale Optimization with a V-shaped transfer function and a 75% accuracy floor constraint, pruning input dimensions by 75.61% (from 41 down to 10 vital features).

• **Spatial-Temporal Classifier:** Combines a 1D Convolutional Neural Network (Conv1D) layer for packet spatial feature extraction with Long Short-Term Memory (LSTM) cells for temporal state tracking.

• **Float16 Edge Quantization:** Applies post-training Float16 weight quantization, compressing model memory footprint to 0.82 MB (83.2% compression) for sub-millisecond execution.

• Real-Time Edge Telemetry to Deep Learning Inference Pipeline:

Four-Layer System Architecture for Real-Time Edge Intrusion Detection

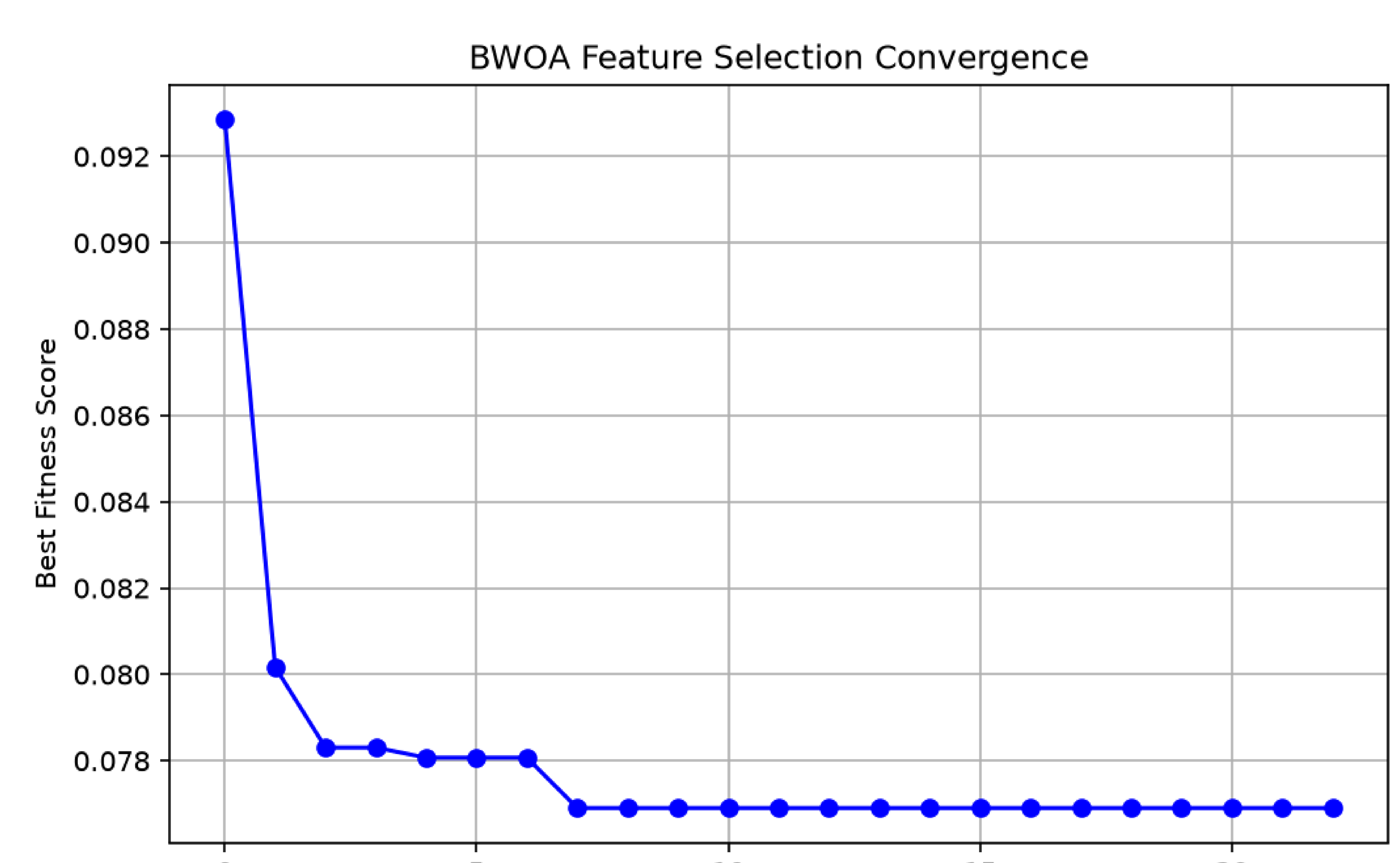


Analysis

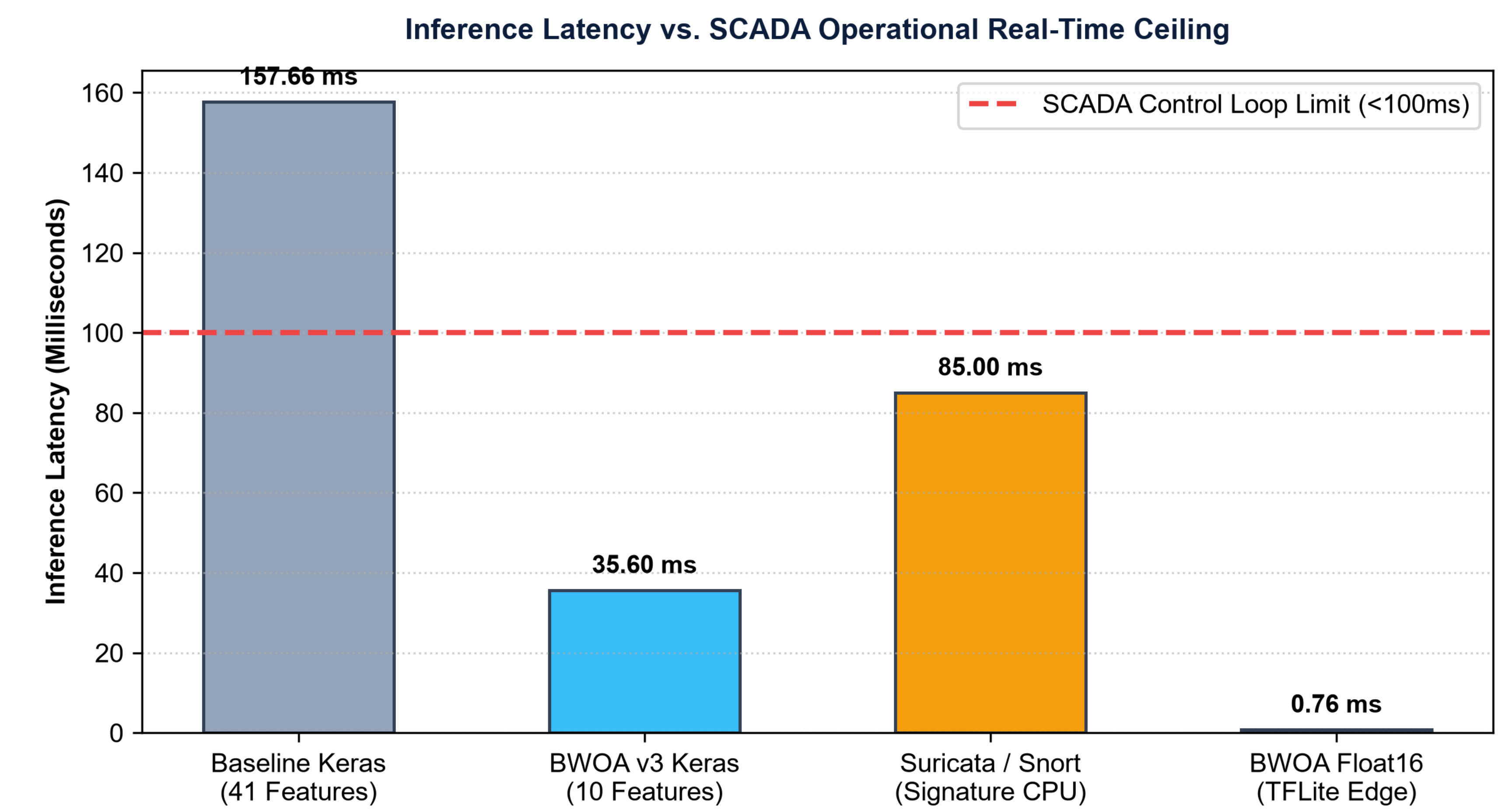
• **BWOA Convergence:** BWOA optimization reached minimal fitness at iteration 23/100, maintaining 92.31% Random Forest cross-validation accuracy on the selected 10-feature subset.

• **Significant Dimension Reduction:** Pruned 31 uninformative features, reducing bandwidth transmission overhead by 75.61% across low-bandwidth satellite links in African extraction sites.

• **Key Discriminative Signals:** Selected features capture connection state (`service`, `flag`), volume asymmetry for DoS (`src_bytes`), privilege escalation (`hot`, `su_attempted`), and error rates (`serror_rate`, `same_srv_rate`).



• Objective latency and multi-class benchmark evaluation on physical edge hardware (Raspberry Pi 4B, 1GB RAM):



Conclusions

• **Sub-Millisecond Edge Real-Time:** Executes single-sample inference in 0.76 milliseconds on a Raspberry Pi 4B (207x faster than baseline), easily satisfying the sub-100ms industrial SCADA control loop deadline.

• **High Detection Fidelity:** Delivers 70.56% multi-class accuracy, 96.89% precision on benign telemetry (preventing false alarms), and 89.04% recall on denial-of-service intrusions.

• **Economic & Life Safety Impact:** Provides 200x to 300x industrial ROI by preventing USD $50,000 to $500,000/hr downtime while safeguarding miner lives from toxic gas or ventilation manipulation (SDG 8 & 9).

• **Open-Source Ecosystem:** Fully open-source CLI agent (@mhiskall282/unesco-mine-sec-cli) and 75/75 automated unit test suites ensure complete reproducibility.

References

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